*.aqu

Purpose

The aquifer group has two input files declared in `type input_aqu` (`src/input_file_module.f90`):

- `aquifer.aqu` defines one record per aquifer object. Each record sets baseflow recession parameters, initial water table depth, initial chemistry, and the recharge and revap controls used by the lumped (non-gwflow) aquifer module. Records are read into `aqudb` (an array of `type aquifer_database`).
- `initial.aqu` is a small table of named initial-condition pointers. Each entry points to organic-mineral, pesticide, pathogen, heavy-metal, and salt initialization files. The `init` column of each row in `aquifer.aqu` selects one of these names.

Source

- Readers: src/aqu_read.f90, src/aqu_read_init.f90
- Type definitions: src/aquifer_module.f90, `type aquifer_database` and `type aquifer_init_data_char`
- Filename declarations: src/input_file_module.f90, `type input_aqu`

aquifer.aqu

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per aquifer with list-directed I/O. The first integer on each record is the aquifer id; the reader uses it as the index into `aqudb`. The remaining columns are read into the fields of `type aquifer_database` in declaration order.

#	Field	Type	Units	Default	Description
1	id	int	none	-	aquifer id. Used as the array index into `aqudb`
2	aqunm	char(16)	none	“”	aquifer name
3	aqu_ini	char(16)	none	“”	initial-condition pointer. Cross-walks with `name` in `initial.aqu`
4	flo	real	mm	0.05	flow from aquifer in the current time step
5	dep_bot	real	m	0.0	depth from mid-slope surface to bottom of aquifer
6	dep_wt	real	m	0.0	initial depth from mid-slope surface to water table
7	no3	real	ppm N	0.0	initial nitrate-N concentration in aquifer
8	minp	real	ppm P	0.0	initial mineral phosphorus concentration in aquifer
9	cbn	real	percent	0.5	initial organic carbon in aquifer
10	flo_dist	real	m	50.0	average flow distance to stream or downstream object
11	bf_max	real	mm	0.0	maximum daily baseflow when all channels are contributing
12	alpha	real	1/day	0.0	lag factor for the groundwater recession curve
13	revap_co	real	none	0.0	revap coefficient. `evap = pet * revap_co`
14	seep	real	fraction	0.0	fraction of recharge that seeps from the aquifer to the deep store
15	spyld	real	$m^{3/m}3$	0.0	specific yield of the aquifer
16	hlife_n	real	days	30.0	half-life of nitrogen in groundwater
17	flo_min	real	m	0.0	water table depth above which return flow occurs
18	revap_min	real	m	0.0	water table depth above which revap occurs

Example

`refdata/Osu_1hru/aquifer.aqu`:

```
aquifer.aqu: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                          init        gw_flo       dep_bot        dep_wt         no3_n
       1  aqu011                     initaqu1       0.05000      10.00000       6.00000       0.00000
       2  aqu012                     initaqu1       0.05000      10.00000       6.00000       0.00000
      21  aqu_deep010                initaqu1       0.00000      10.00000      20.00000       0.00000
```

Notes:

- The tool header names (`gw_flo`, `no3_n`, `sol_p`, `carbon`, `alpha_bf`, `revap`, `rchg_dp`, `spec_yld`, `hl_no3n`) differ from the source field names (`flo`, `no3`, `minp`, `cbn`, `alpha`, `revap_co`, `seep`, `spyld`, `hlife_n`). Column order is what matters.
- The `init` column points to `initaqu1`, which must exist as a row in `initial.aqu`.
- In this dataset all shallow aquifers share the same property values. The last record (`aqu_deep010`) represents the deep aquifer with no baseflow (`flo = 0`, `flo_min = 0`, `revap_min = 0`).

initial.aqu

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per initial-condition set with list-directed I/O. Records are read into `aqu_init_dat_c` (an array of `type aquifer_init_data_char`).

#	Field	Type	Description
1	name	char(16)	initial set name. Cross-walks with `aqu_ini` in `aquifer.aqu`
2	org_min	char(16)	name of organic-mineral initial-state record
3	pest	char(16)	name of pesticide initial-state record
4	path	char(16)	name of pathogen initial-state record
5	hmet	char(16)	name of heavy-metal initial-state record
6	salt	char(16)	name of salt initial-state record

The literal `null` is accepted in slots that have no initial-state file.

Example

`refdata/Osu_1hru/initial.aqu`:

```
initial.aqu: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                        org_min             pest             path             hmet             salt
initaqu1                    no_init             null             null             null             null
```

Notes:

- The tool writes a trailing `description` column. The source type has no `description` field. List-directed reads consume tokens left to right, so the extra trailing token is harmless. Do not rely on the description being read.
- `no_init` is itself a name that must resolve in the organic-mineral initial file. `null` is treated as "no file".

Related files

- aquifer.con. Connectivity entries that map each aquifer id to its spatial location and downstream object.
- `aqu_catunit.ele` and `aqu_catunit.def`. Catchment-unit elements that group aquifers.
- Organic-mineral, pesticide, pathogen, heavy-metal, and salt initial-state files referenced by `initial.aqu`.